

soils4med

SOIL health monitoring and information systems FOR
sustainable soil management in the MEDiterranean region

MedSIS manual

Data Managers

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DATASETS



MAPS



DOCUMENTS



GEOSTORIES



DASHBOARDS

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1. Overview

This manual has been developed as a reference guide to assist Med-SIS STAFF or Data Managers in navigating and using the Med Soil Information System (Med SIS).

The SIS is essentially a **Web GIS Catalogue** of resources about soil data. It is based on Geonode (<https://geonode.org/>), whose resources are represented by datasets, maps, documents, geostories and dashboards.

In addition to the standard features of the catalog for managing **soil data**, it was necessary to add an additional tool—known in MED SIS as the "**Back Office**."

MED-SIS staff and data managers generate catalog resources in the back-office and manage them within the Web GIS catalog.

So the document is divided in three main sections: "Web GIS Catalogue", "Back Office", "Admin panel"

Further details on Web GIS Catalogue functionalities can be found on the official GeoNode pages, at the link <https://docs.geonode.org/en/latest> by clicking on the "User Guide" menu

2. Web GIS Catalogue

Within the SIS, only a limited subset of content is publicly accessible. Registered generic users are therefore restricted to read-only access and can interact with resources only in accordance with the permissions explicitly defined by the data manager responsible for each dataset.

The data manager is responsible for data ingestion into the Med-SIS and **defines the access control policies** applied to each dataset, **based on the intended users and their roles**. He is the one that adds datasets or documents, creates maps, dashboards or geostory, edits and shares resources with other users.

Med-SIS homepage

The MEd-SIS system is locked by default; this means that the user must log in to view the homepage.

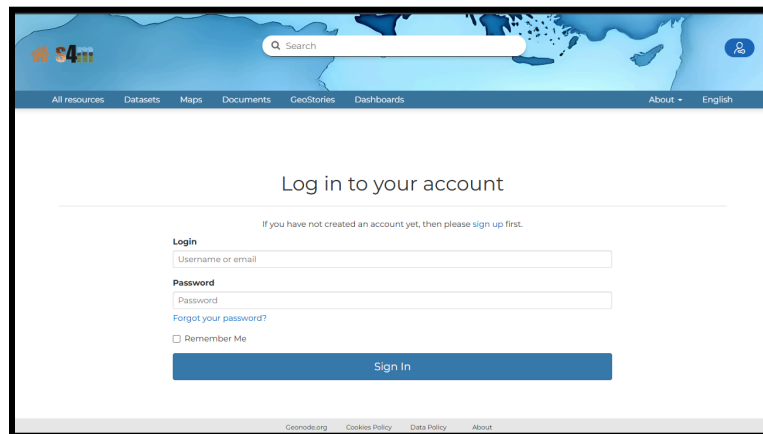
The screenshot shows the login page of the Med-SIS system. At the top, there is a navigation bar with a search bar and a user profile icon. Below the navigation bar, the main heading is "Log in to your account". A subtext says "If you have not created an account yet, then please [sign up](#) first." The login form includes fields for "Username or email" and "Password", a link for "Forgot your password?", and a checkbox for "Remember Me". A blue "Sign In" button is at the bottom of the form. The footer contains links for "Contact Us", "Cookies Policy", "Data Policy", and "About".

Figure. 1 login page

Registration (Figure 2) is handled by the SIS staff; that is, it must be validated in the **Admin panel** to become effective. Once registered, the user will be able to access the catalog.

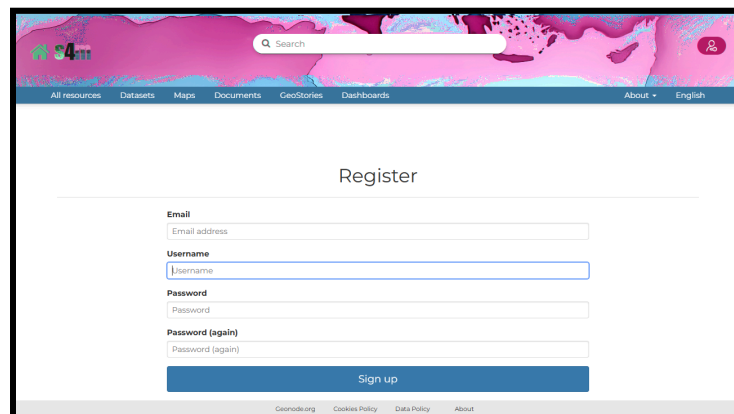
The screenshot shows the registration page of the Med-SIS system. At the top, there is a navigation bar with a search bar and a user profile icon. Below the navigation bar, the main heading is "Register". The registration form includes fields for "Email address", "Username", "Password", and "Password (again)". A blue "Sign up" button is at the bottom of the form. The footer contains links for "Contact Us", "Cookies Policy", "Data Policy", and "About".

Figure. 2 registration page



Figure. 3 Med-SIS homepage after the login

As shown in Figure 2 the system's main page displays the project logo in the upper-left corner, which redirects to the homepage when clicked.

In the upper-right corner there are: the language selection menu and the icon for personalized user access showing user-related actions, illustrated in Figure 3.

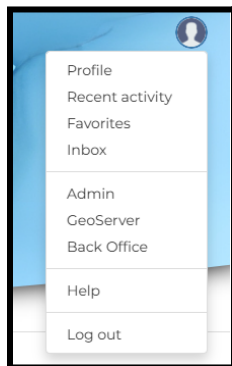


Figure 4

- Profile: it shows the user profile
- Recent activity: it shows recent user activities
- Favorites: it shows the list for favorites datasets
- Inbox: it shows the messages sent to the user by other SIS users
- Admin: it go to the admin panel if the user is in the Med-SIS staff
- Geoserver: open the Geoserver application
- Back office: open the back office if the user is in the Med-SIS staff
- Help: it shows a quick help to do action in the catalogue
- Log out

In the top left of the SIS homepage, there are the menus for the direct access to the available resources, organized by category (Datasets, Maps, Documents, Geostories, and Dashboards) or all together (All Resources).

They can also be accessed by scrolling down the page.

In the center of the page, there are quick links that provide direct access to the following resources:

- **Soil Maps** : Vector / Raster datasets with Pedological Classification
- **Soil Point Data** : Point vector datasets with profile and samples soil data
- **Chemical Health** : Vector / Raster datasets with soil indicators describing chemical health
- **Physical Health**: Vector / Raster datasets with soil indicators describing physical health



Figure. 5 Resources' List and Filters

Once the list of available resources is displayed (Figure. 5);, they can be filtered with one or more of the following filters:

tags. Several Med-SIS keywords was added (e.g “Pedological map”)

- **Regions:** the region filter is also based on metadata, but it refers specifically to the geographical region associated with the resource (it has nothing to do with bounding box which is automatically derived from the data extent)
- **Owner:** it filters resources based on the user that created or owns them
- **Group:** it filters resources that are associated with a specific GeoNode group
- **Date from:** resources based on their creation or upload date, showing only resources added on or after a specified date
- **Date to:** resources newer than the selected date
- **Extent:** resources whose geographical area intersects or is contained within the selected extent

User Profile

Once registered, the SIS administrator will assign new members the Data Manager role in accordance with the system’s role-based access control policies.

Communication among the different actors within the SIS can take place through the platform’s internal messaging system or via private email.

By selecting the “Profile” option from the user menu, the data manager is directed to the profile management interface, where

- configuring personal information by clicking the *Edit Profile* button, where users can update their profile by adding a profile photo, specifying their organization, writing a description, indicating their position and setting their location
- managing email settings by clicking the *Associates e-mails* button
- managing credentials (e.g., password) by clicking the *Set/Change password* button
- accessing a summary of the personal activities in the SIS by clicking the *My Activities* button that displays the user’s latest actions within the system
- setting the modes on how, when, and for which events receiving notifications about activities on the platform by other users or administrators by clicking the *Notification* button
- creating announcements
- inviting new users to join the SIS platform.

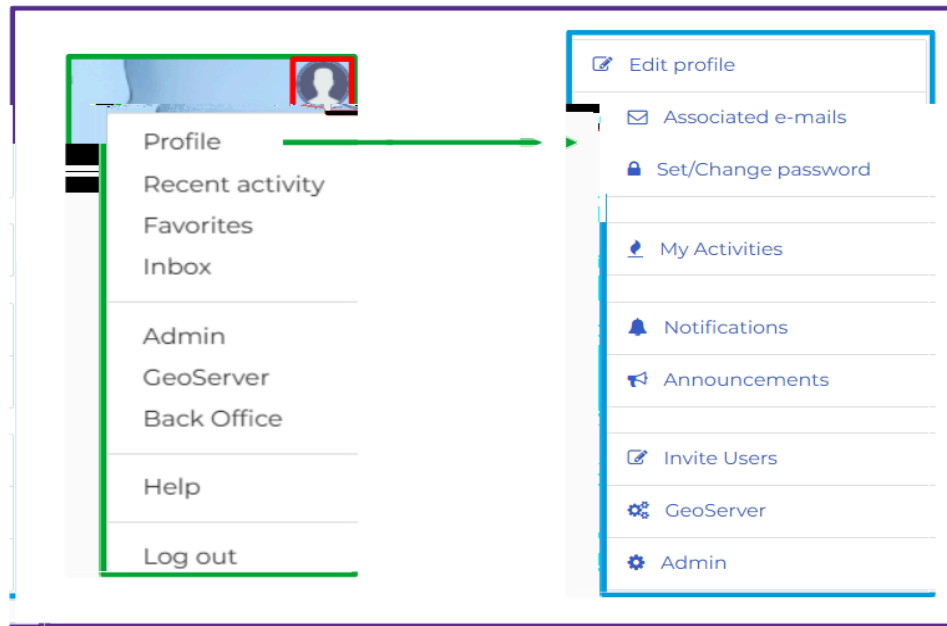


Figure 5 User profile menus

The **Recent Activity** option provides a detailed overview of the actions performed within the SIS from all the active users, allowing to track and to view the type of activities, and to identify the users who performed them.

The **Favorites** option accesses the list of the resources marked as the user's *favorites* allowing them to easily retrieve the resources without the need of searching for them in the catalog.

The **Inbox** option is part of the Geonode internal messaging system, designed for communication between registered users and group members. It acts as a hub to receive, read, and reply to messages from the SIS platform.

The other options of the menu include access to administrative tools such as **Admin** and **Geoserver**, the **Back Office** for uploading and managing soil data, **Help** section that provides guidance on using GeoNode functionalities and **Log out** button.

Admin Panel and *Back Office* will be described in respective sections, while Geoserver, responsible for publishing and serving the spatial data of the SIS platform through standard OGC web services (such as WMS, WFS, and WCS), It should not be used by data managers, but only by the administration.

< Go back to Point Soil Data Availability

Attribute filter

Match **all** of the following conditions:

surveyors	=	Barron	
elev_m_asl	>	8	

- **Load style:** opens the tool to load an SLD style, provided the user has "Edit" or "Manage" permissions. Familiarity with the Style Layer Descriptor (SLD) standard is recommended.
- **Map layer setting:** set layer display options for maps
- **Edit Metadata:** edit metadata of the dataset, very important in filtering datasets
- **Upload Metadata:** upload a XML file with dataset metadata (use with cautions)
- **Update Dataset:** change data uploading a new file (use with cautions)

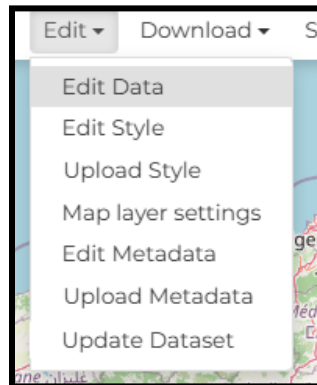


Figure 8: Edit Menu

by clicking on the **Download menu**, the user can: download the dataset as a compressed **.zip** file containing the corresponding shapefile, which can be opened using GIS software; export the dataset in different formats (GeoJSON, GML-FeatureCollection, GML-GFS 1.1.0 FeatureCollection, Shapefile, CSV). Users can also export just the dataset displayed in the current view (Crop dataset to current viewport) or the subset defined by the applied filters (Download filtered dataset). The filter type "viewport" means that search results are limited to resources that fall within the current map extent (the visible area on the screen); download the ISO metadata in .xml format

Figure 9: Download form

By clicking the **Share menu**, the user sets permissions for the dataset, provided they have "Manage" permissions. Using the permissions settings widget, the user can define dataset permissions for other users. The permissions can be "View", "Download", "Edit", or "Manage".

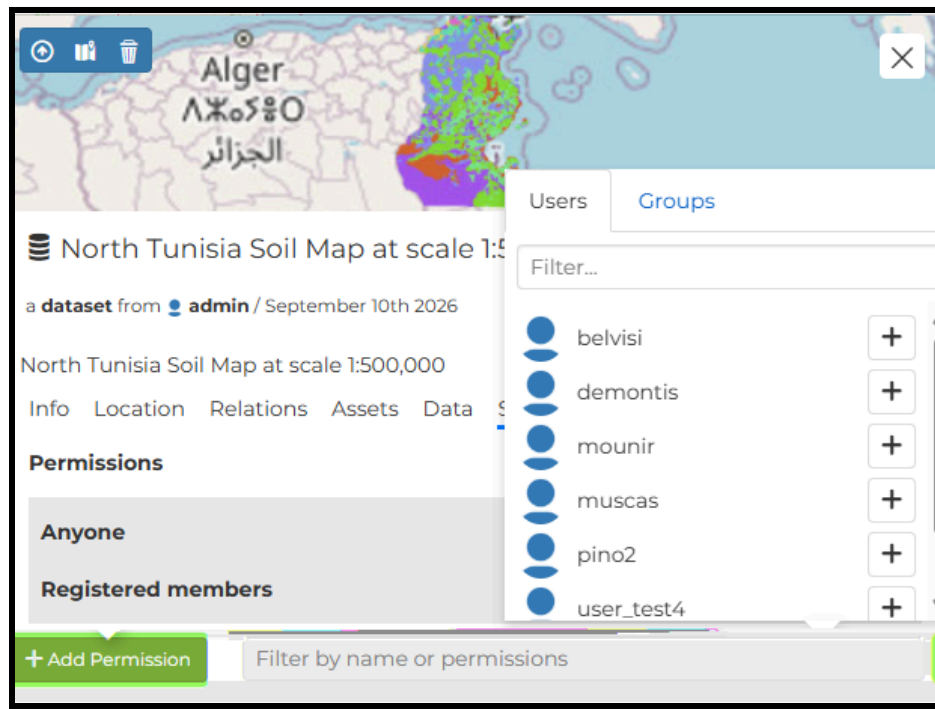


Figure 10: Permissions settings widget

To protect the dataset, it is recommended to set the option to "none" for anonymous and registered users. Furthermore, it is more secure to assign permissions to individual users rather than to groups

by clicking on the **Filter menu** user can perform the same operations described for the icons of the **View menu**. The user is allowed to perform advanced searches (attribute-based filtering, and area of interest selection). As for the attribute filtering, it is possible to perform multiple filtering adding new conditions with the "+" button.

by clicking on the **Create Map link** the user can create a new **Map resource** using the dataset

Maps

Maps are compositions or visualizations built from *one or more datasets* (layers).

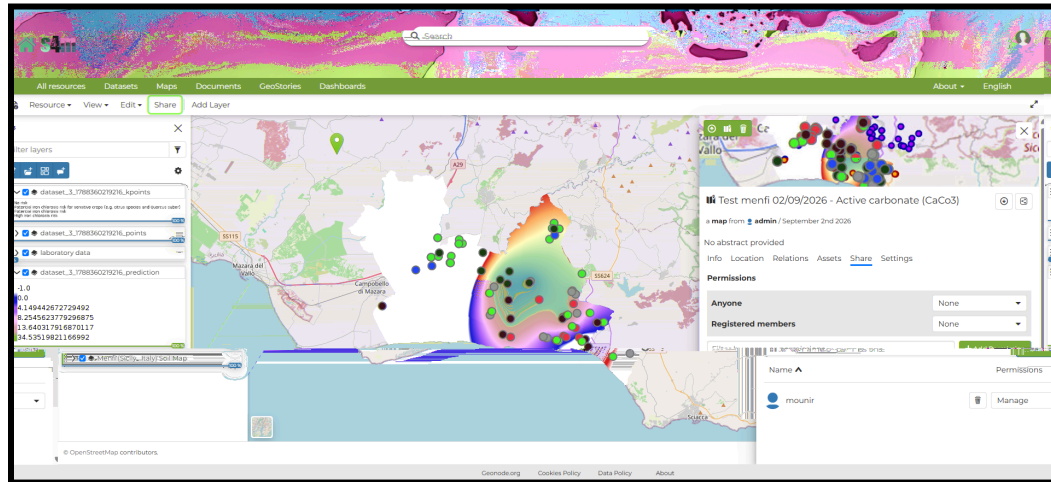


Figure 11, Map resource panel



Figure 12 Map Resource menu

If the user has permissions to view the maps and their associated datasets, they can:

by clicking on the **Open Property button**, the user can contemplate the resource's general info, geographic location, alphanumeric attributes, and the linked resources that use it or are used by it

by clicking on the **Resource menu** a user with staff permissions can save or delete the resource

by clicking on the **View menu**, the user can contemplate the resource's metadata:

by clicking on the **Edit menu**, the user can edit the metadata of the map, very important in filtering datasets or add widget (It will be necessary to configure some widgets)

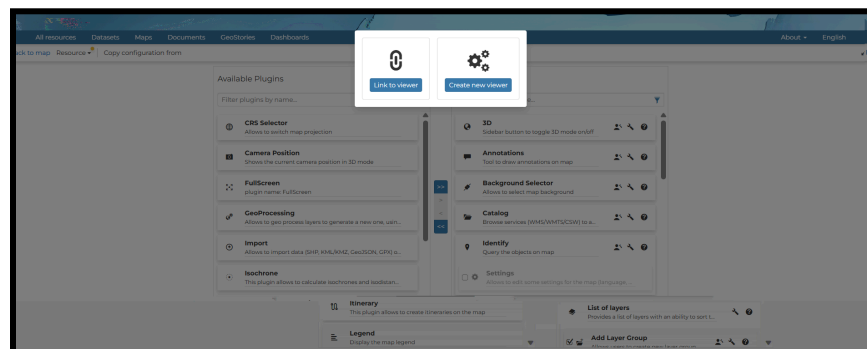


Figure 13, add widget panel

By clicking the **Share menu**, the user sets permissions for the dataset, provided they have "Manage" permissions. Using the permissions settings widget, the user can define dataset permissions for other users. The permissions can be "View", "Download", "Edit", or "Manage".

By clicking the **Add Layer link**, the user can add datasets to the map. The datasets to be added to the map can be selected from a list of catalog or external datasets. External datasets must be configured by clicking the "+" button in the interface.

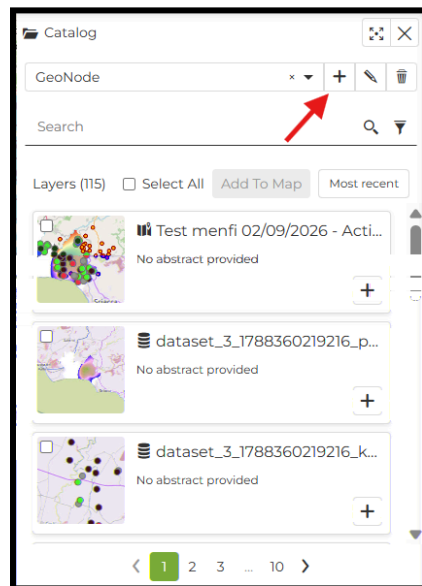


Figure 14, add Layer panel

The panel to explore the map configuration is shown in Figure 15

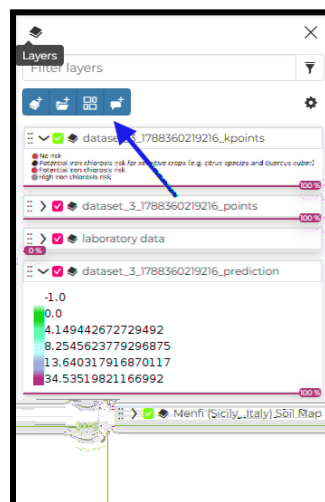


Figure 15, Map properties panel

By clicking the **first** of the four buttons on the left, the data managers can add external layer to import choosing between WMS (Web Map Service), WMTS (Web Map Tile Service), CSW (Catalog Service for the Web), TMS (Tile Map Service), XYZ Tiles, WFS (Web Feature Service) and ArcGIS REST

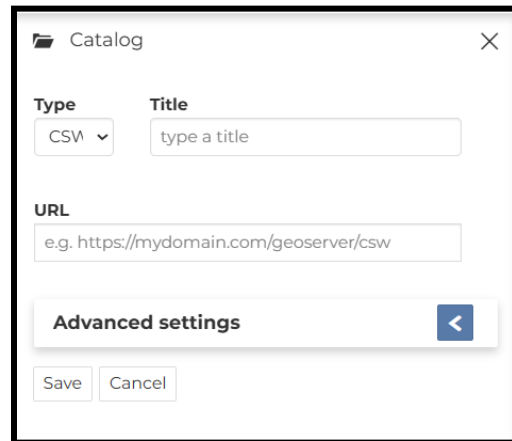
The image shows a 'Catalog' dialog box with a close button (X) in the top right corner. It contains two columns: 'Type' and 'Title'. The 'Type' column has a dropdown menu currently set to 'CSW'. The 'Title' column has a text input field with the placeholder text 'type a title'. Below these columns is a 'URL' section with a text input field containing the example 'e.g. https://mydomain.com/geoserver/csw'. At the bottom of the dialog is an 'Advanced settings' section with a blue arrow icon pointing left. At the very bottom are 'Save' and 'Cancel' buttons.

Figure 16, Add layer panel

The Advanced settings control how a remote layer is:

Loaded: service search (automatically retrieves available layers from the selected remote service), server type (defines the type of remote server, e.g., GeoServer, ArcGIS, to ensure proper requests)

Displayed: preview (displays a preview of the layer before adding it), format (sets the image format, e.g., PNG, JPEG, affecting quality and transparency), single tile vs tiled (requests the map as one large image instead of multiple tiles: better for quality, slower performance), tile size (sets the size of tiles, e.g., 256×256, impacting performance and number of requests)

Optimized: visibility limits (controls at which zoom levels the layer is visible), tile grids (uses the tiling scheme defined by the remote service for better alignment), domain aliases (specifies alternative domains for the same service: useful for load balancing or browser limits)

Interacted with: interactive legend (allows users to interact with the legend, e.g., toggle styles or classes), information sheet (defines the format of feature info shown when clicking on the map, e.g., HTML, JSON).

By clicking the **second** of the four buttons on the left, users can add a new group of features to the current map.

By clicking the **third** and **fourth** of the four buttons on the left, users can add annotations and widgets to the current map: users can typically add: Points (markers), Lines, Polygons, Text labels / notes. Users can customize the annotation setting color, style (line thickness, fill, icon), label or description. If users have edit permissions, the annotations become part of the map. For a generic user, annotations are usually temporary and lost when refreshing.

Documents

Documents are used to attach and share non-spatial files uploaded to the platform, such as: PDFs (reports, studies), Word documents, Excel spreadsheets, Images, Presentations.

Users can: open documents (e.g., PDF preview) and download them (depending on permissions). Like all the other SIS other resources, documents may have the options to be opened, downloaded, edited and managed but SIS generic users can usually only:

- **View** the document and its metadata
- **Download** the document and the ISO metadata

Geostories

GeoStories are a special content type used to create interactive narratives combining maps and multimedia content. GeoStories are essentially web-based storytelling pages that integrate: maps already present in the SIS, text (descriptions, explanations), images and media Embedded content.

If the user has permissions to view the resources in the geostory, they can:

- **view** the story and its metadata
- **scroll** through sections/slides
- **read** text and see media
- **interact** with embedded maps
- **explore** geographic content
- **follow** the narrative flow navigating between sections (slides/blocks)

Dashboards

Dashboards are interactive pages used to visualize and analyze geospatial data through charts, widgets, and maps. Dashboards are data visualization tools that combine: Maps, Charts (bar, pie, line, etc.), Indicators /counters, Filters and widgets with the goal to summarize and explore data in a dynamic, visual way.

If the user has permissions to view the resources in the geostory, they can:

- **View** the dashboard and its metadata (charts, maps, and widgets)
- **Interact** with visualizations
- **Apply filters** and see dynamic changes based on filters
- **Explore** linked maps

If the user has permissions to edit and manage, they can add widget

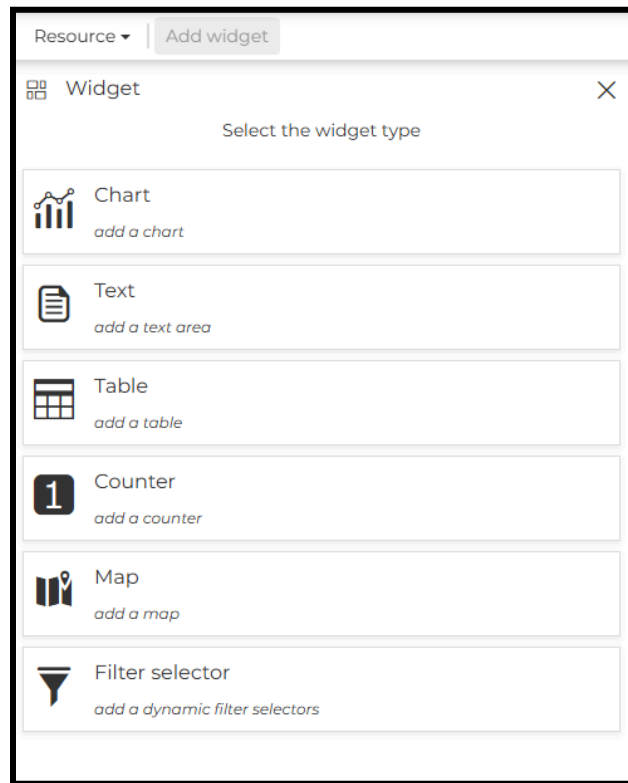


Figure 17, Add widget panel

3. Back Office

In addition to the standard features of the catalog for managing **soil data**, it was necessary to add an additional tool—known in MED SIS as the "**Back Office**."

Soil data is managed in the MED-SIS Back-office and published in the MED-SIS Catalogue

The Back Office is accessible only to data managers and SIS staff

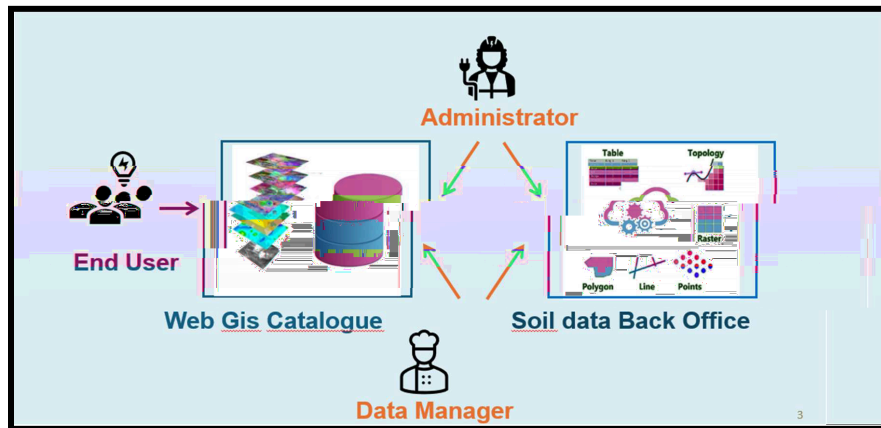


Figure 18, MED-SIS schema

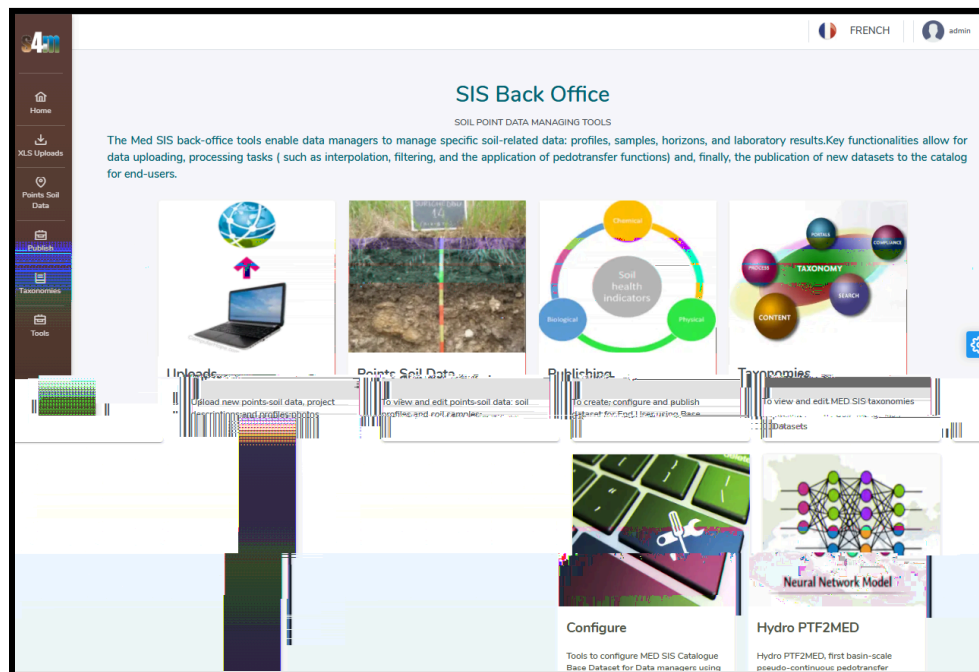


Figure 19, Back Office home page

Data Managers use Back Office tools to manage, filter, and process **base datasets** in order to create and publish **customized datasets for end users**.

The Back Office shows in the first page (Figure 19) the available tools accessible clicking on the respective description card:

Configure : This is the first tool to use after installation. The tool configures the **base datasets** and prepares them to the catalogue

Upload : The tool permits to upload new Soil Data and Metadata from XLSx files: point soil data; projects' metadata, profiles' photos metadata, Laboratory Extra Measure data

Point Soil Data : The tool permits to view and edit data for soil profiles or soil samples

Taxonomies : The tool describes all the taxonomies used for point-based soil data.

Publishing : The tool enables the preparation, processing, and validation of datasets for end users.

- **Hydro PTF2MED** : The tool calculates a basin-scale pseudo-continuous pedotransfer function specifically developed for the Mediterranean region.

Configure

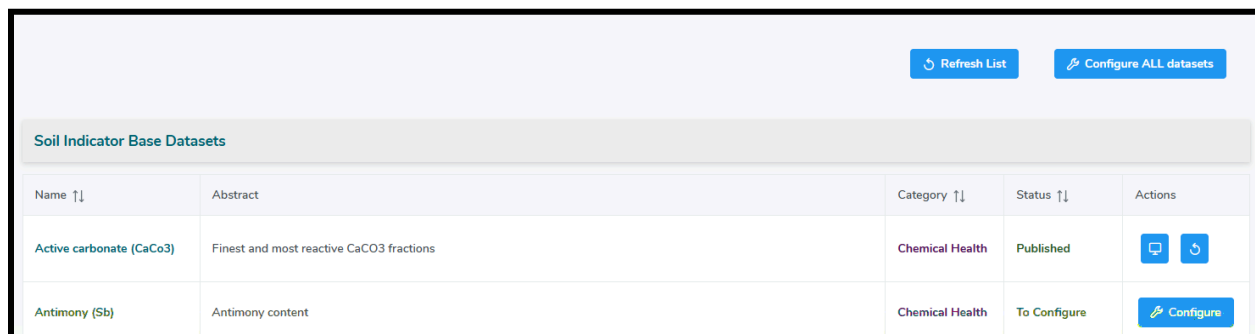
Following installation, the MED SIS requires the configuration and publication of "**base datasets**" which enable the automatic updating of soil data whenever new data is uploaded.

*The "**base datasets**" should be accessible only to data managers and SIS Staff*

The list contains all available **base datasets** (soil indicators and point-based soil data) that can be configured using the "Configure" or "Configure all datasets" buttons.

The configuration is performed on the server and may take a few minutes.

The command may cause data access issues during the update; use with caution.



Soil Indicator Base Datasets				
Name ↑↓	Abstract	Category ↑↓	Status ↑↓	Actions
Active carbonate (CaCo3)	Finest and most reactive CaCO3 fractions	Chemical Health	Published	 
Antimony (Sb)	Antimony content	Chemical Health	To Configure	

Figure 20, the Configure tool

Upload

Data managers use the MED-SIS Back office to store soil data in the database. The primary soil data to be uploaded are point-based soil data, which include information on profiles and samples. Other informations needed by MED-SIS are: Project's metadata, Photo's metadata and laboratory extra measures that are measures not implemented in the default data template.

The format of uploaded data is the **XLSx format**, a template can be found on the doc folder in the Github repository (https://github.com/crs4/med_sis/tree/main/src/s4m_catalogue/static/doc) for each type of information.

Specifically, point-based soil data are managed as catalogue resources, form part of the "**base datasets**" and drive the automatic generation of the soil indicators that constitute the other part.

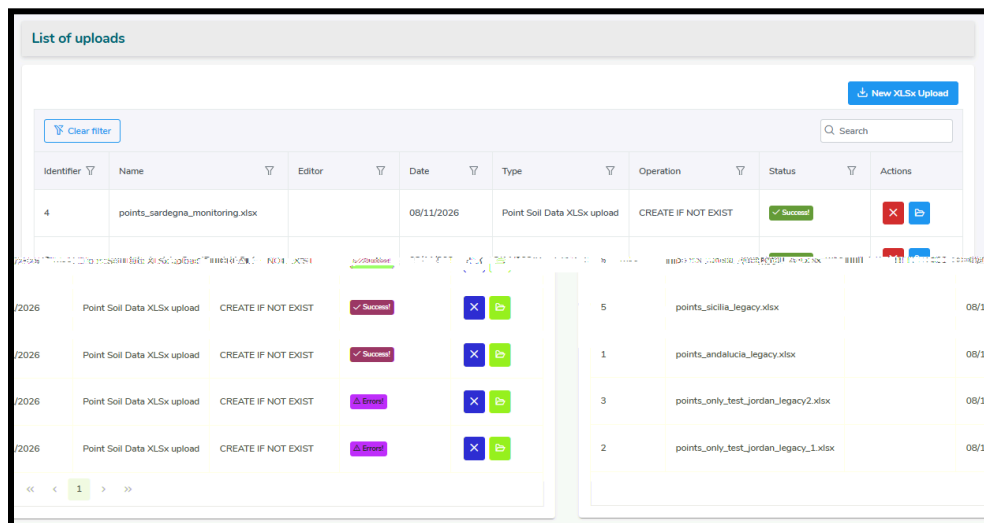
For these activities, the process typically starts from the Excel file complaint with the templates available for download from the back office.

The *XLS Uploads* tool button in the back office home page opens the *List* of uploads that have already been uploaded into the SIS with all the related details (file name, person who performed the action, date of uploading, file type and action performed) and its outcome (Status).

By clicking the  icon the user delete the list element but **NOT THE RELATIVE IMPORTED DATA !!!!!**

By clicking the  icon the user opens the upload metadata panel

By clicking the **"New XLSx upload"** the user open the **"New XLSx upload "** panel



Identifier	Name	Editor	Date	Type	Operation	Status	Actions
4	points_sardegna_monitoring.xlsx		08/11/2026	Point Soil Data XLSx upload	CREATE IF NOT EXIST	Success	 
2026	Point Soil Data XLSx upload	CREATE IF NOT EXIST	Success	 	5	points_sicilia_legacy.xlsx	08/11
2026	Point Soil Data XLSx upload	CREATE IF NOT EXIST	Success	 	1	points_andalucia_legacy.xlsx	08/11
2026	Point Soil Data XLSx upload	CREATE IF NOT EXIST	Error	 	3	points_only_test_jordan_legacy2.xlsx	08/11
2026	Point Soil Data XLSx upload	CREATE IF NOT EXIST	Error	 	2	points_only_test_jordan_legacy_1.xlsx	08/11

Figure 21 List of uploaded dataset

An Upload has four possible states:

- **ELABORATING:** The upload is being processed
- **SUCCESS:** The dataset has been uploaded in the server,
- **ERRORS:** The dataset has been uploaded in the server with some errors,
- **CRITICAL ERROR :** The upload generate system errors

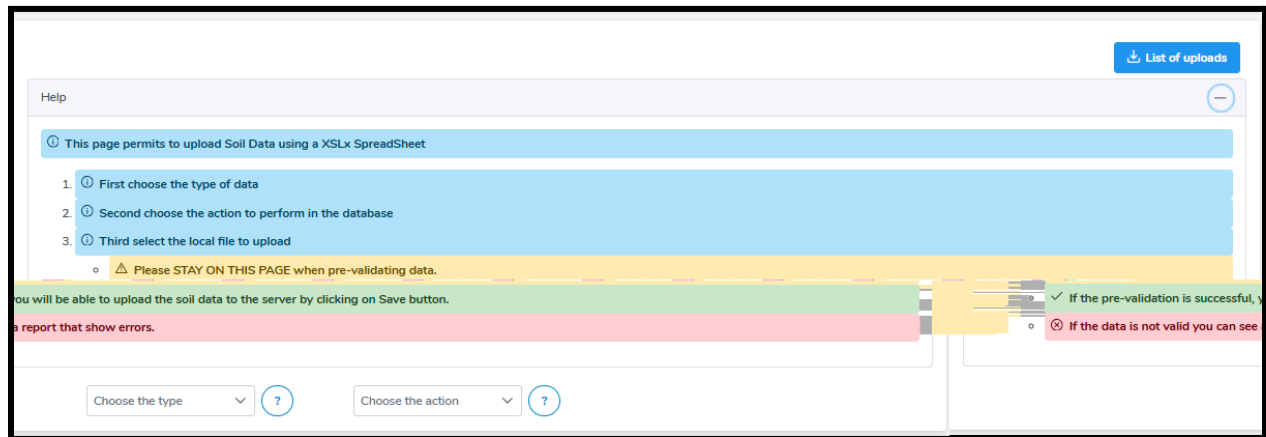


Figure 22 New XLSx upload panel

The *New XLSx upload panel* display several help sections directly or showed by clicking the  icon

The data manager selects the option for data type: Point Soil Data; Laboratory Extra Measure; Project Metadata; and Photo Metadata and action. Then the option for the database action:

- **CREATE IF NOT EXIST** (For each item in the upload the SIS tries to write it in the database. If an item with the same id already exists, it throws an error.
- **REPLACE IF EXIST:** For each item uploaded, the SIS attempts to replace it with the one in the database. If the item doesn't exist an error is generated.
- **UPDATE IF EXIST:** For each item uploaded, the SIS makes partial changes to the fields of the item in the database. If an item with the same id doesn't exist, an error is generated.

For each type, users can click the question mark next to the options list to download the official .XLS template and the instructions for filling it out. Once the options have been selected, the "Choose file" button appears.

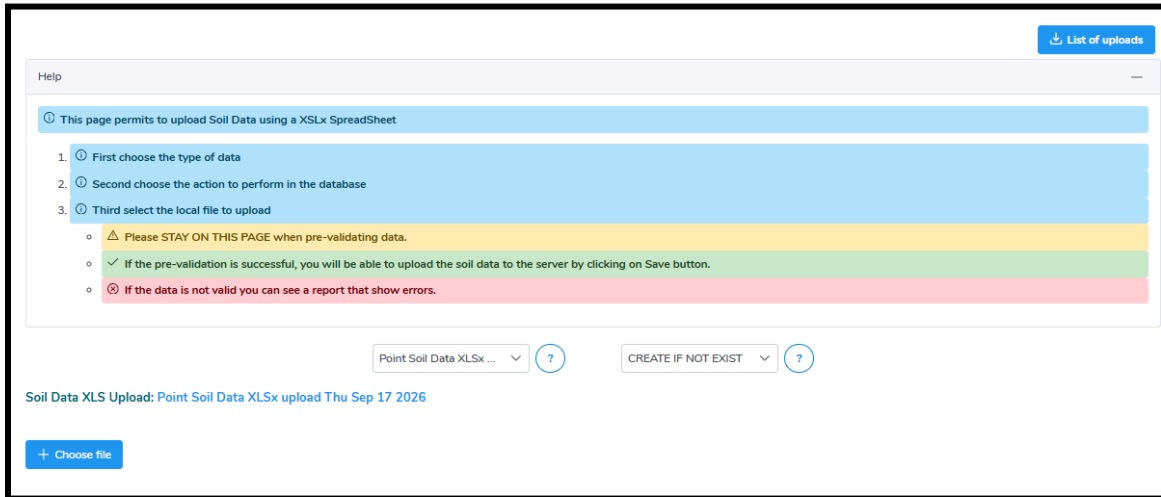


Figure 23, Upload tool page

Once the XLS file is uploaded successfully, a confirmation message indicating that the file has been successfully converted to a GeoJSON format will appear. Users can scroll down to view the uploaded points on a map. If no errors occurred during the upload, a No Errors message will appear below the map, as shown in Figure 8.

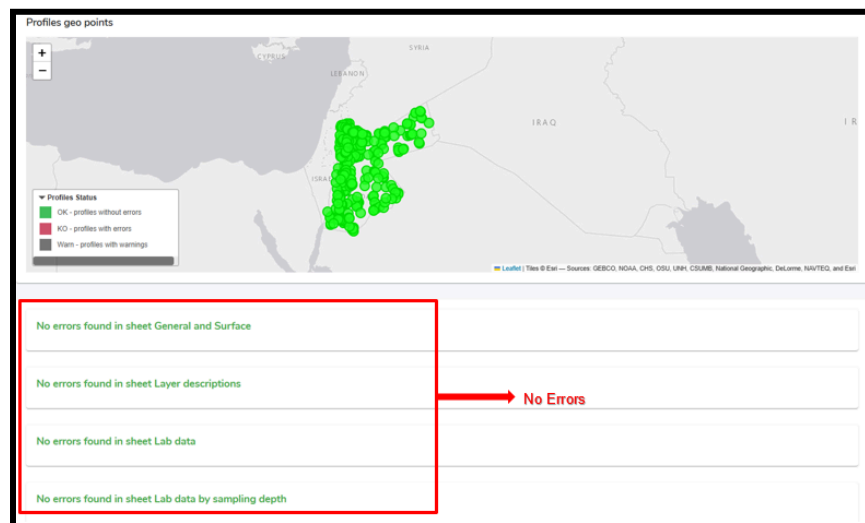


Figure 24 Dataset chosen and errors preview

Immediately after, by clicking the **Reset** button, users can restart the upload process while by clicking **Save Data**, the system will be uploading the data to Database. This process may take several minutes



Figure 25 Dataset chosen and “Save Data” button

If errors occurred during the upload is possible to view a simple log reporting the type of error and the cell (row and column) in the XLSx file



Figure 25 Dataset chosen with errors

Sheet "Lab data": 9Errors!			
▼ Collapse			
Element ↑↓	Row ↑↓	Column ↑↓	Error ↑↓
?	111	1 or A	Error: duplicate key found
?	114	1 or A	Error: duplicate key found
?	115	1 or A	Error: duplicate key found
?	116	1 or A	Error: duplicate key found
MED3100@1	463	58 or BF	Wrong Percentage value field
TPER001@2	482	13 or M	Wrong Percentage value field
TPER001@2	482	17 or Q	Wrong Percentage value field
?	483	1 or A	Error: duplicate key found
CARBSKB004@1	537	58 or BF	Wrong Percentage value field
<< < 1 > >> 10 ▼			

Figure 26 Errors log

Important instructions for uploading **Point soil data** Excel files to the SIS

As in the XLS template file *MED-SIS-TEMPLATE_POINT_SOIL_DATA.xlsx* the spreadsheet must have four sheets:

- 1. General and Surface**
- 2. Layer description**
- 3. Lab data**
- 4. Soil Classification**

The data in sheets 1 and 2 must start from line 10 and line 6 respectively, and the data in sheets 3 and 4 must start from line 3.

Before uploading the file, ensure that the data in Excel is in 'value' format and that no fields have been filled in with formulas or text formatting rule; otherwise, SIS will return errors:

Error msg	Description
Wrong Classification code field	The classification code in the cell is unknown by the SIS, <i>Change it</i>
Wrong Number field	The value in the cell isn't a number <i>Check if the cell contains a formula instead a value</i>

Wrong Boolean field	The value in the cell isn't a Boolean. Allowed values T for True, F for False' <i>Check if the cell contains a formula instead a value</i>
Wrong Date field	The value in the cell isn't a Date. Allowed format is DD/MM/YYYY <i>Check if the cell contains a formula instead a value</i>
Wrong Percentage value field	The value in the cell isn't a valid percentage. <i>Check if the cell contains a formula instead a value</i>
Wrong Positive value field	The value in the cell should be a positive number or 0. <i>Check if the cell contains a formula instead a value</i>
Wrong Latitude value field	The latitude should be in decimal degrees. <i>Check if the cell contains a formula instead a value</i>
Wrong Longitude value field	The longitude should be in decimal degrees. <i>Check if the cell contains a formula instead a value</i>
Error found Rich Text instead of Plain Text	<i>There are Text Formatting Rules</i>
Error: duplicate key found	<i>The element identifier is the same as another element in a previous row</i>

Point Soil Data

The tool permits CRUD (Create, Read, Update) operations on a single soil profile or sample. As shown in the Figure xxx the first page of the tool shows the list of profiles and samples and some elements that the user can use to search a specific element in the list

Figure 27 Point Soil Data list

By clicking the  icon the user delete the list element (**AND THE RELATIVE IMPORTED DATA !!!!!**)

By clicking the  icon the user opens the **Point Soil Data panel** for the element

By clicking the  the user can add a new element in the list

By clicking the  icon the user can edit the field value

By clicking the  icon the user opens the information panel for the field

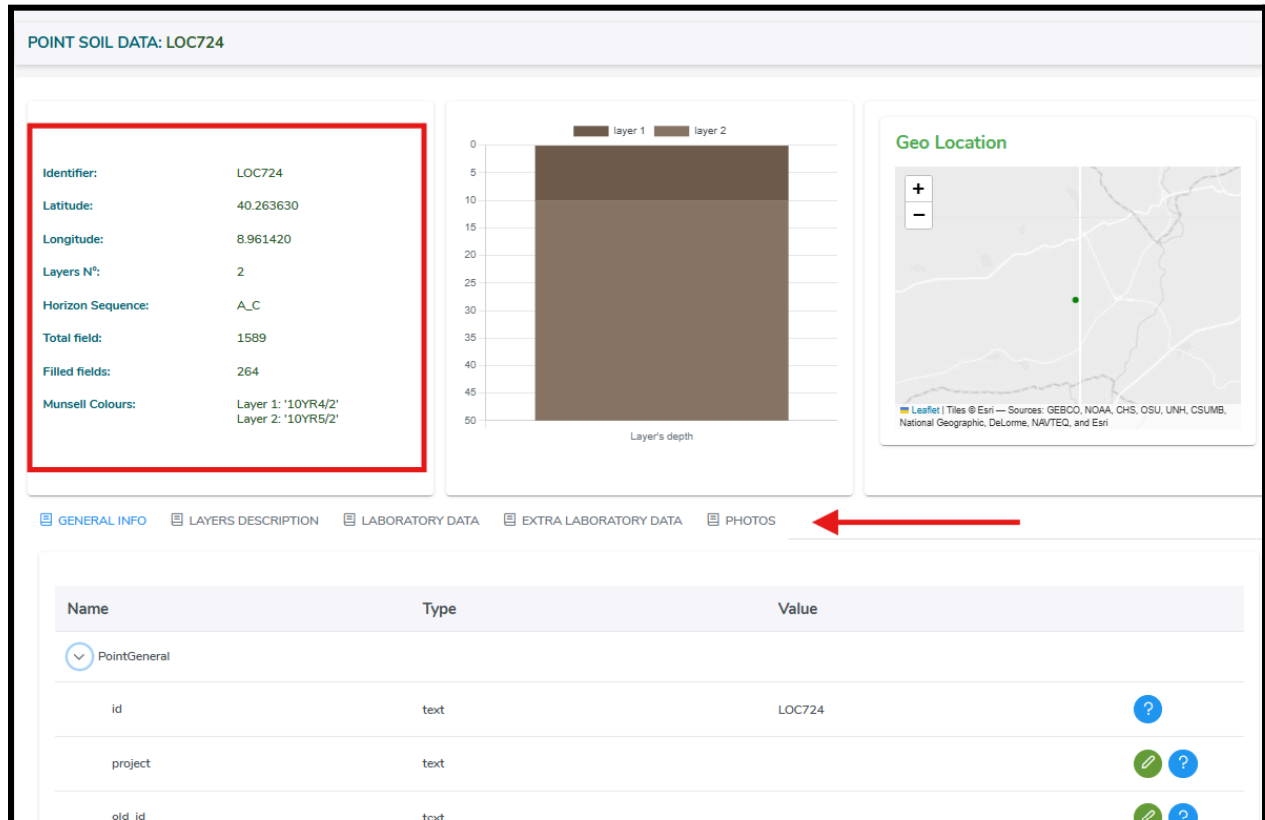


Figure 28 Point Soil Data view

Taxonomies

The Taxonomies tool allows you to explore all the taxonomies used in the creation of soil data.

The screenshot shows the 'Taxonomies' tool interface. At the top, there is a dropdown menu with 'ABUNDANCE_PARTICLES_TYPES' selected. Below the dropdown, there is a section titled 'TAXONOMY:ABUNDANCE_PARTICLES_TYPESList of values'. Under this title, there is a descriptive text: 'Classes of abundance of volcanic glasses in the sand and coarse silt fraction, see table 8.79 WRB 4th edition 2022'. Below the text is a table with four columns: 'Identifier', 'Value', 'Description', and 'default.URI'. The table contains five rows of data.

Identifier	Value	Description	default.URI
ABUNDANCE_PARTICLES_TYPES:C	C	class: Common; description: > 5 - 30	
ABUNDANCE_PARTICLES_TYPES:F	F	class: Few; description: > 0 - 5	
ABUNDANCE_PARTICLES_TYPES:M	M	class: Many; description: > 30	
ABUNDANCE_PARTICLES_TYPES:N	N	class: None; description: 0	
ABUNDANCE_PARTICLES_TYPES:ND	ND	class: no data; description: no data	

Figure 29 Taxonomy list

The drop-down list allows the user to select the taxonomy that will then be displayed as a list. Each item in the list will describe a taxonomy entry using: the “**identifier**” used in the database; the “**Value**” or entry’s name; the “**description**” of the entry and an external “**URI**” of an international standard vocabulary (e.g <https://qudt.org/vocab/unit/MilliGM-PER-KiloGM>) .

By Using the  , the user can open the panel to create a new taxonomy, which may be necessary for entering additional laboratory measurements.

These operations must be coordinated with other information systems (SIS) to ensure **interoperability** and data sharing within the network.

The screenshot shows the 'Taxonomy create' form. It has two main sections: 'Identifier, use uppercase characters A-Z, 0-9, _ (underscore) without spaces' and 'Description'. The 'Identifier' section has a text input field with the value 'TEST_NEW_TAXONOMY'. The 'Description' section has a text input field with the value 'Description of the new taxonomy'. At the bottom of the form, there are two buttons: 'Cancel' and 'Save Changes'.

Figure 30 Taxonomy create

After creation, new entries can be added.

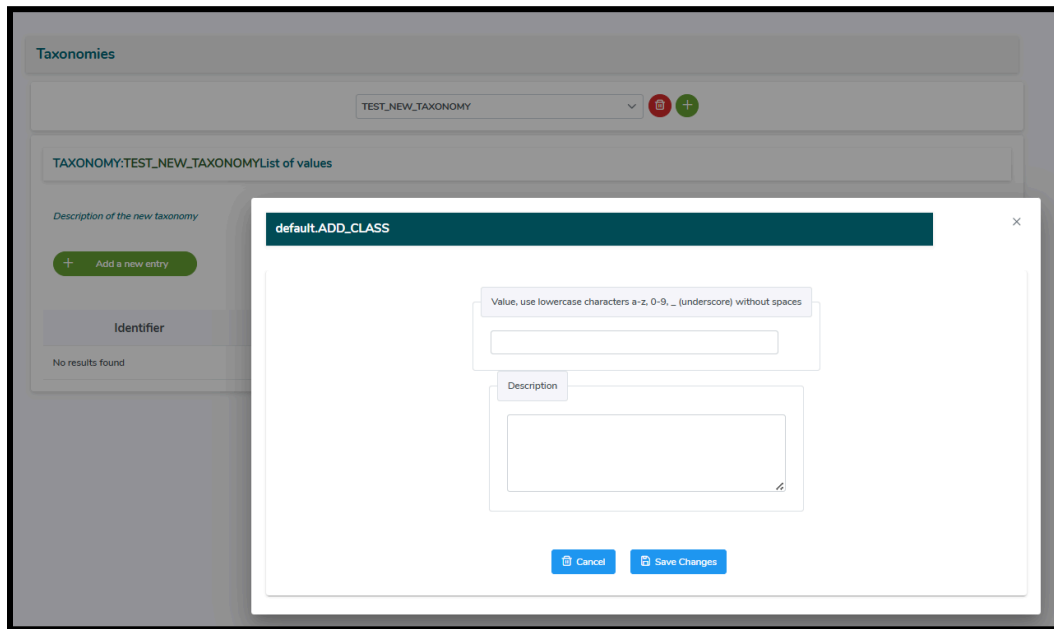


Figure 31 Taxonomy - add classification

It is also possible to delete the taxonomy using the  icon.

To ensure interoperability, **default taxonomies cannot be removed or modified.**

Publishing

The *Publishing* tool button in the back office home page opens the **main tool** of the MED SIS that **permits to create customized datasets for end user**

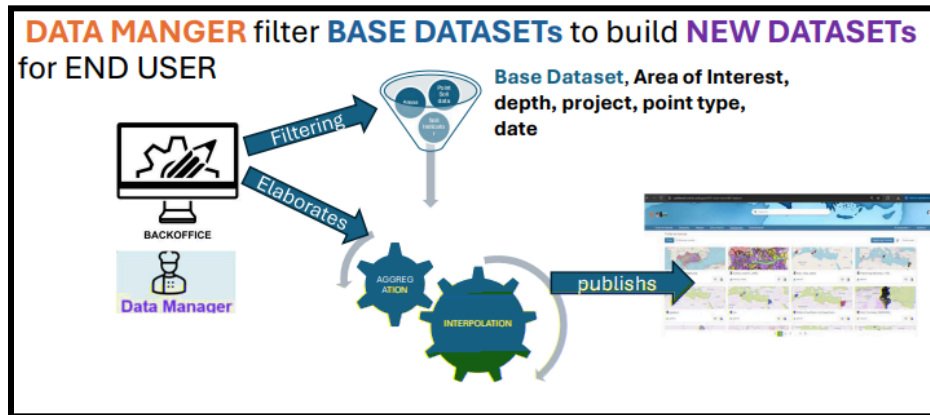


Figure 32 Publishing data flow

The tool in the first page shows the *list* of datasets that have already been created with all the related details and some icons for possible actions to perform.

Dataset List									
+ New Dataset									
Clear Search									
Actions	Code ↑↓	Name ↑↓	Date ↑↓	Context ↑↓	Source ↑↓	User ↑↓	Email ↑↓	Status ↑↓	
  	4	admin-dataset-2026-09-10	2026-09-10	p Soil Indicator	Cation Exchange Capacity (CEC) / Clay ratio	admin	demontis@crs4.it	CREATED	
  	3	admin-dataset-2026-08-31	2026-08-31	p Soil Indicator	Active carbonate (CaCo3)	admin	demontis@crs4.it	PUBLISHED	
<< < 1 > >>									

Figure 33 Publishing - customized dataset list

By clicking the  icon the user delete the list element (**but NOT THE RELATIVE DATASET IN THE CATALOGUE !!!!!**)

By clicking the  icon the user opens the dataset panel that show the details

By clicking the **“New Dataset”** the user open the **“New Dataset”** panel (Figure 32)

There are three type of dataset depending on the source base dataset type:

- Point Soil Data Sections: Using a Base Dataset source of type Point Soil Data
- Soil Indicators: Using a Base Dataset source of type Soil Indicator
- Soil indicators evaluated in an area: Using a Base Dataset source of type Soil Indicator

When created the dataset appears in the list of dataset with the status “CREATED”

Choose the context of the new Dataset

You must choose the context of the new dataset

☐ Point Soil Data Sections
☐ Soil Indicators
☐ Soil indicators evaluated in an area

Figure 34 Publishing - customized dataset types

Dataset List									
+ New Dataset									
Clear Search									
Actions	Code	Name	Date	Context	Source	User	Email	Status	
Edit Delete	4	admin-dataset-2026-09-10	2026-09-10	Soil Indicator	Cation Exchange Capacity (CEC) / Clay ratio	admin	demontis@crs4.it	CREATED	
Edit Delete	3	admin-dataset-2026-08-31	2026-08-31	Soil Indicator	Active carbonate (CaCo3)	admin	demontis@crs4.it	PUBLISHED	
Edit Delete	6	admin-dataset-2026-09-17	2026-09-17	Soil Indicator		admin	demontis@crs4.it	CREATED	
Edit Delete	7	admin-dataset-2026-09-17	2026-09-17	Soil Indicator		admin	demontis@crs4.it	CREATED	
<< < 1 > >>									

Figure 35 Publishing - customized dataset types

A dataset has four possible states:

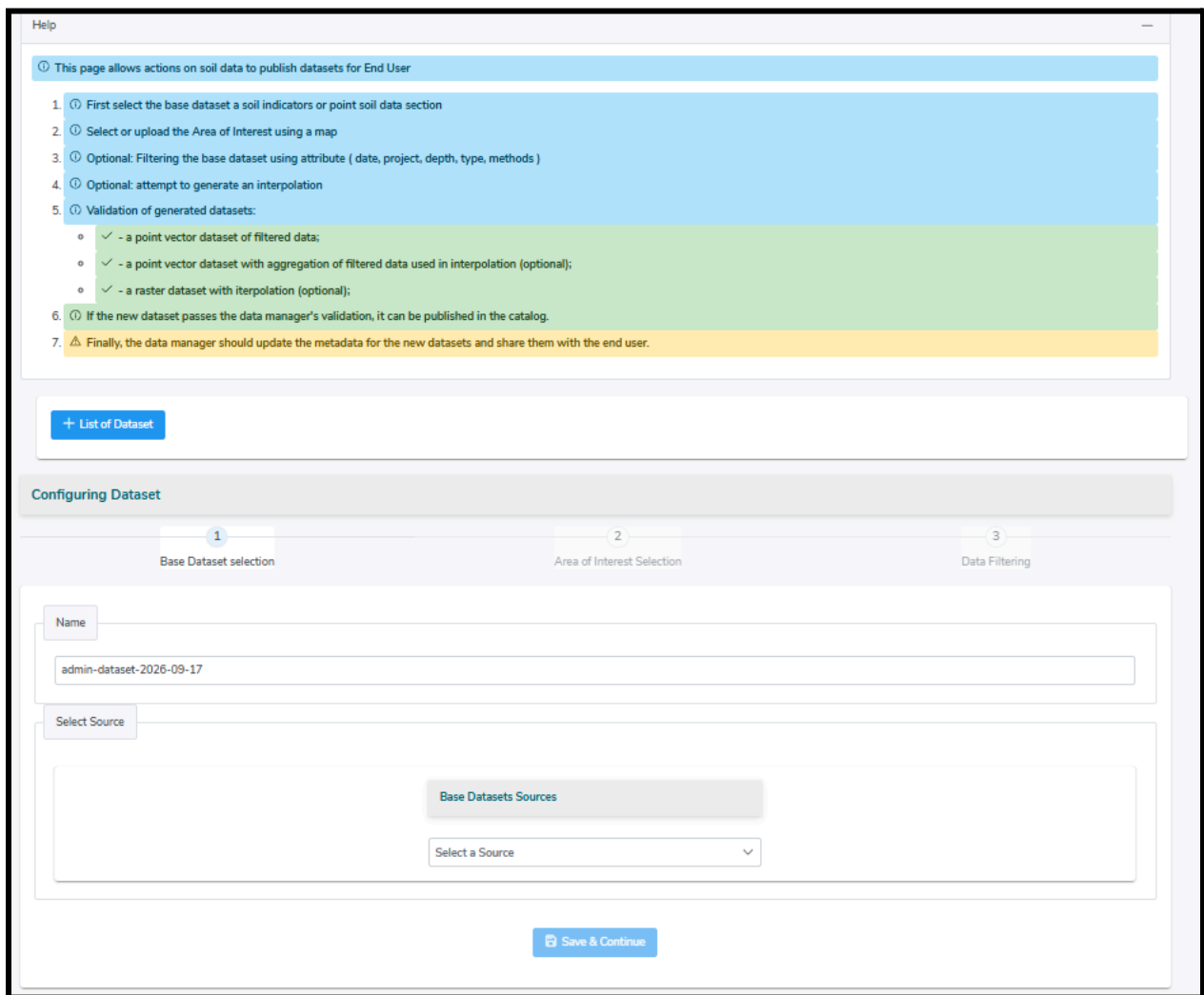
- **CREATED** : The dataset has been created but not configured
- **CONFIGURED** : The dataset has been configured but not published in the catalog

- **IN_PROCESS** : The dataset is being processed
- **VALIDATED** : The dataset has been validated by the data manger,
- **PUBLISHED** : The dataset has been published in the catalogue,
- **ERRORS** : The elaboration of the dataset generate some errors

By using the tool the data manger change the state of the dataset data manager from “CREATED” to “PUBLISHED”:

CREATED => CONFIGURED => IN_PROCESS => VALIDATED => PUBLISHED

By clicking the  icon the user opens the dataset panel that show the status of the datasets and



Help

① This page allows actions on soil data to publish datasets for End User

- ① First select the base dataset a soil indicators or point soil data section
- ① Select or upload the Area of Interest using a map
- ① Optional: Filtering the base dataset using attribute (date, project, depth, type, methods)
- ① Optional: attempt to generate an interpolation
- ① Validation of generated datasets:
 - ✓ - a point vector dataset of filtered data;
 - ✓ - a point vector dataset with aggregation of filtered data used in interpolation (optional);
 - ✓ - a raster dataset with interpolation (optional);
- ① If the new dataset passes the data manager's validation, it can be published in the catalog.
- ⚠ Finally, the data manager should update the metadata for the new datasets and share them with the end user.

+ List of Dataset

Configuring Dataset

1 Base Dataset selection 2 Area of Interest Selection 3 Data Filtering

Name

admin-dataset-2026-09-17

Select Source

Base Datasets Sources

Select a Source

Save & Continue

Figure 36 Publishing - dataset configuration

Configuring Dataset

1 Base Dataset selection 2 Area of Interest Selection 3 Data Filtering

Name
admin-dataset-2026-09-17

Select Source

- Sodium Adsorption Ratio (SAR) sodicity
- Sodium Adsorption Ratio (SAR) toxicity
- Sodium Exchangeable Percentage (ESP) sodicity
- Sodium Exchangeable Percentage (ESP) waterlogging
- Soil Erodibility by Water
- Soil Erodibility by Wind
- Select a Source

Save & Continue

Figure 37 Publishing - Configuration Step 1. Base Dataset Selection

After the selection the system load **ALL** the points in the selected Base Dataset and the user can click to “Save & Continue” button

Configuring Dataset

1 Base Dataset selection 2 Area of Interest Selection 3 Data Filtering

Name
admin-dataset-2026-09-17

Select Source

Selected source: Sodium Exchangeable Percentage (ESP) sodicity

Points: 1202

Sodicity indicator based on exchangeable sodium and cation exchange capacity (CEC) (Ali, 2011)

Save & Continue

Figure 38 Publishing - Configuration Step 1. Base Dataset Selected report

The next step in configuration is the Area of Interest selection that can be performed using a custom polygon (Figure 35) or a Boundaries Catalogue dataset (Figure 36)

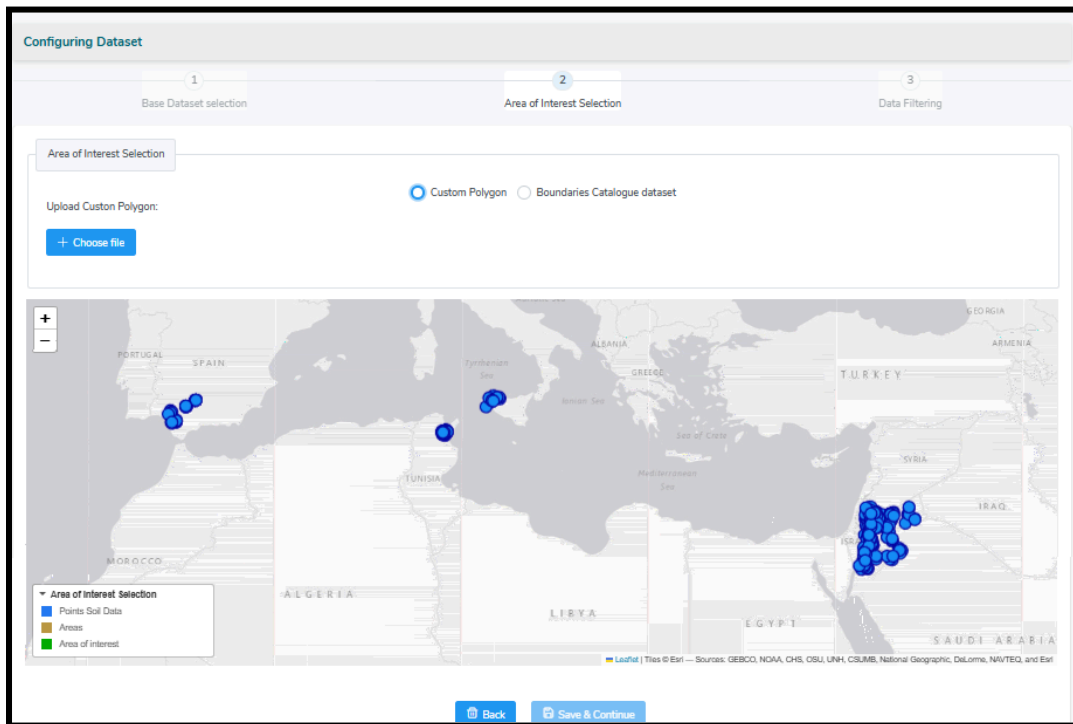


Figure 39 Publishing - Configuration Step 2. Custom polygon as AOI

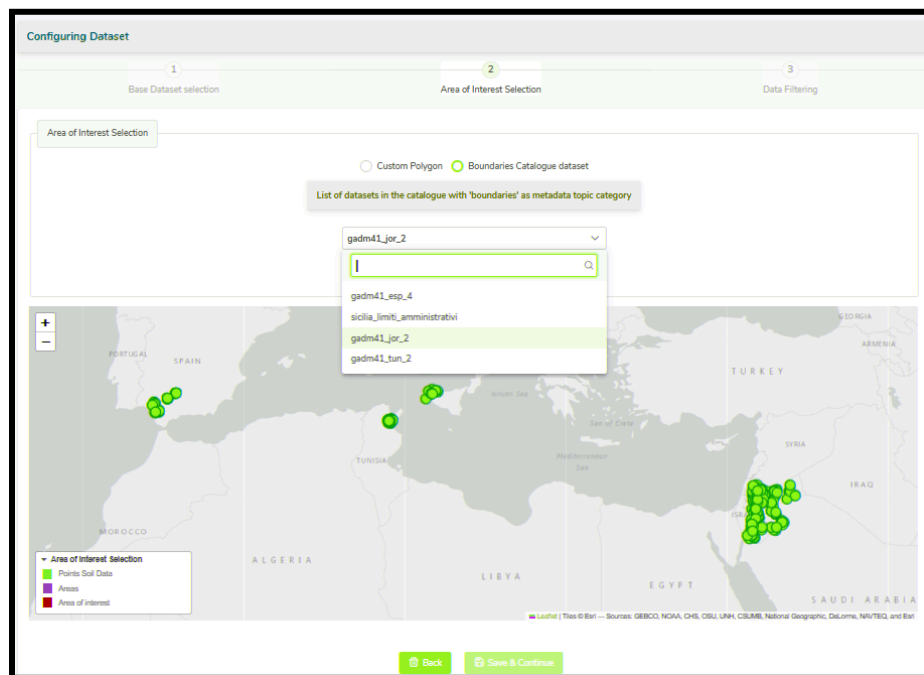


Figure 40 Publishing - Configuration Step 2. Catalogue resources as source for AOI

To be classified as "Boundaries," a dataset in the catalog must have its "Topic category" set to "Boundaries" in the metadata.

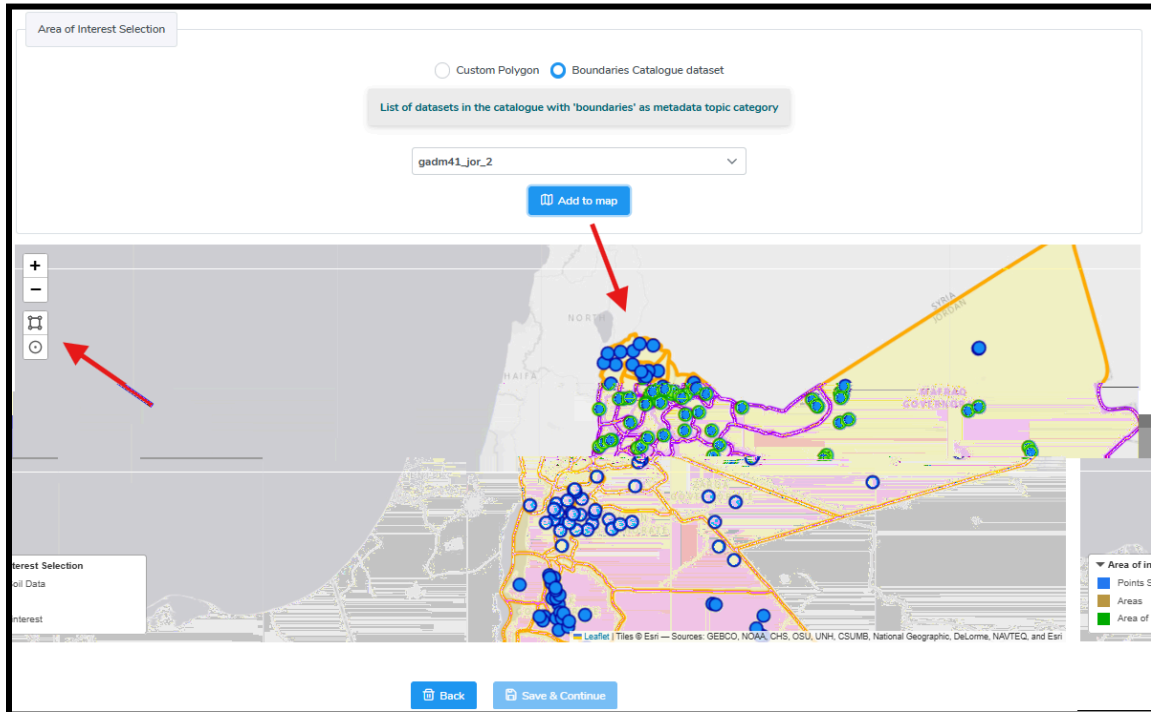


Figure 41 Publishing - Configuration Step 2. Catalogue add source for AOI

After adding the dataset to the map, the user can select polygons using a point or a selection box after clicking the tool located in the top-left corner of the map.

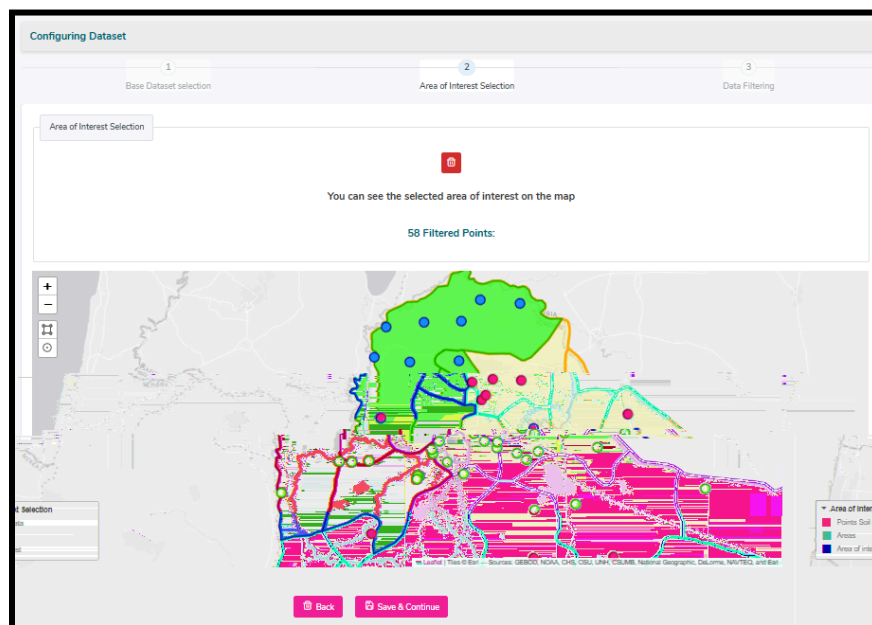


Figure 42 Publishing - Configuration Step 2. AOI selection

After the AOI selection the number of filtered points is shown in the page and the user is able to click on the “Save & Continue” button

The next step in configuration is filtering some attributes of points:

- The **period of observations** (start and end dates) or all dates
- The **type** : *Auger hole*, *Minipit* or *Profile* or all types
- The Identifier or name for the **project**, or survey activity/campaign, in the frame of which the points soil data data were generated or all projects
- The **depth** (upper and lower depth if present). The user can select 0 to 20 cm, 20 to 50 cm or all depths
- The **method used in laboratory** for the measure, none or all methods (it depends on involved measures - only for soil indicators)
- The **method used to carry out the field survey** or all methods:

Identifier	Code	Description
SURVEY_METHODS:CP	CP	method: CPCS;
SURVEY_METHODS:FA	FA	method: FAO
SURVEY_METHODS:FA06	FA06	method: FAO; year: 2006
SURVEY_METHODS:FA52	FA52	method: FAO; year: 1952
SURVEY_METHODS:FA77	FA77	method: FAO; year: 1977
SURVEY_METHODS:FA90	FA90	method: FAO; year: 1990
SURVEY_METHODS:US	US	method: USDA soil survey manual
SURVEY_METHODS:US18	US18	method: USDA soil survey manual ; year: 2018
SURVEY_METHODS:US52	US52	method: USDA soil survey manual ; year: 1952
SURVEY_METHODS:US62	US62	method: USDA soil survey manual ; year: 1962
SURVEY_METHODS:US75	US75	method: USDA soil survey manual ; year: 1975
SURVEY_METHODS:US93	US93	method: USDA soil survey manual ; year: 1993
SURVEY_METHODS:WR	WR	method: WRB;
SURVEY_METHODS:WR22	WR22	method: WRB; year: 2022

Figure 43 Survey Methods

The filtering panel (Figure 41) shows a form to filtering and a map to visualize the result

The page in the top indicates the number of points selected and of the aggregated point needed in the interpolation. The aggregation uses the location (10m point buffer) and a media of weighted values based on depth intervals (only for soil indicators).

All the value in drop down lists in the form refers to the content of the selected points.

After the filtering the user is able to click on the “Save & Continue” button

58 Filtered Points

17 Aggregated Points

Apply the filter

Filter for all periods

From date

01/01/1970

To date

01/01/1970

Upper: 0 cm; Lower: 20 cm

☐

Upper: 20 cm; Lower: 50 cm

☐

Type or All types

Method or All methods

Projects

Method or All methods

Back

Save configuration

Points Filtering

Point Soil Class

Area of Interest

Filtering by date

Select a period or None

Filtering by depth

All depths

Upper

Type

Select a point soil data

select a type

Survey method

Select a survey method

select a survey method

Project

Select a project or All

select a project

Laboratory method

Select a laboratory method

select a method

Figure 44 Publishing - Configuration - Filtering panel

Now the customized dataset is on validation's phase.

The system displays a report.

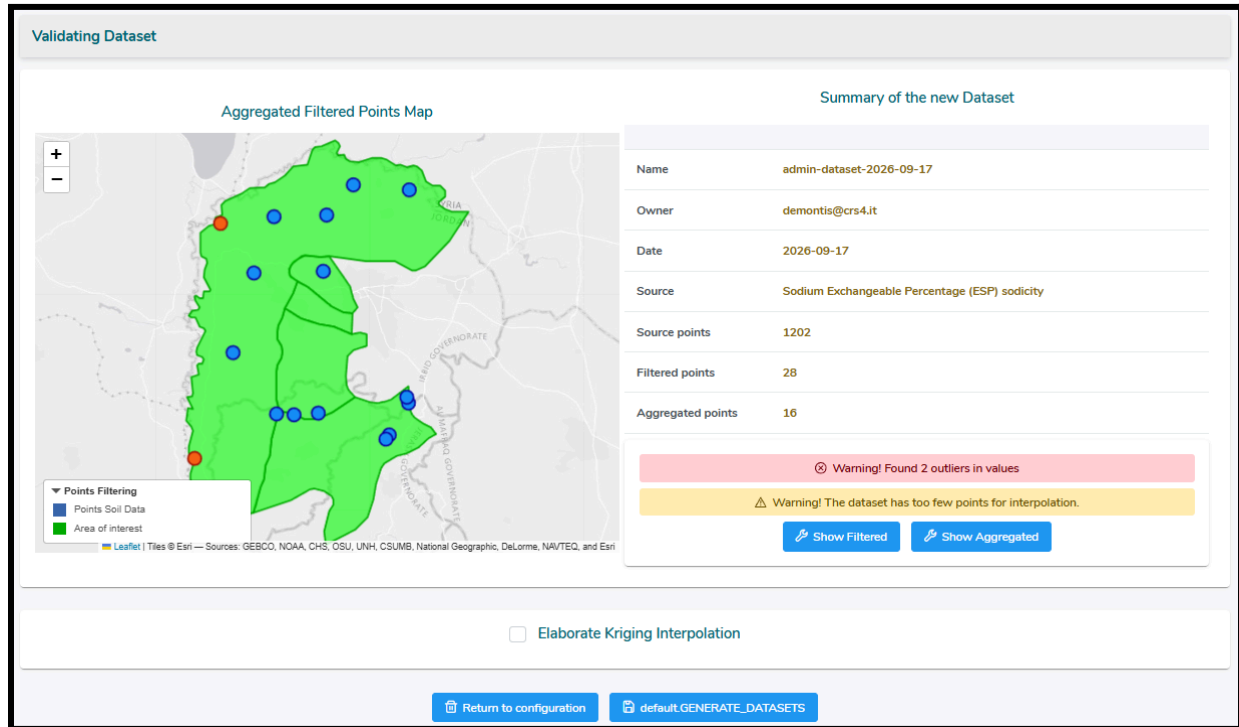


Figure 45 Publishing - Validation - Report panel

The report indicates the numbers of filtered and aggregated points and if the dataset contains **outliers**.

The presence of outliers generates out of scale predictions by skewing parameter estimates and **altering the underlying data distribution**.

Outliers can result from data entry errors, but they can also be genuine data points that WARRANT MORE INVESTIGATION.

If the dataset does not require further operations, such as Kriging interpolation, it can be published in the catalog.

Otherwise, selecting the "Elaborate Kriging Interpolation" checkbox opens the Kriging interpolation panel (Figure 43).

Using the panel is possible to set the interpolation parameters and evaluate the resulting semi variogram (Figure 44)

If the validation of the results is successful, the data manager will be able to generate the new datasets.

Kriging Interpolation

Select Kriging interpolation parameters

☐ Do not use outliers

The presence of outliers generates out of scale predictions by skewing parameter estimates and altering the underlying data distribution. Outliers can result from data entry errors, but they can also be genuine data points that WARRANT MORE INVESTIGATION.

Model (Default: linear " $a + x * b$ ")

Linear

$a + b * x$

Initial Number of Distance Classes (Minimum: 2, Default: 100)

-

100

+

Maximum Distance Floating point (Default: -1.000000, not used)

-

-1

+

Skip Number (Minimum: 1, Default: 1)

-

1

+

Calculate the variogram

Return to configuration

default.GENERATE_DATASETS

Figure 46 Publishing - Validation - Kriging Interpolation

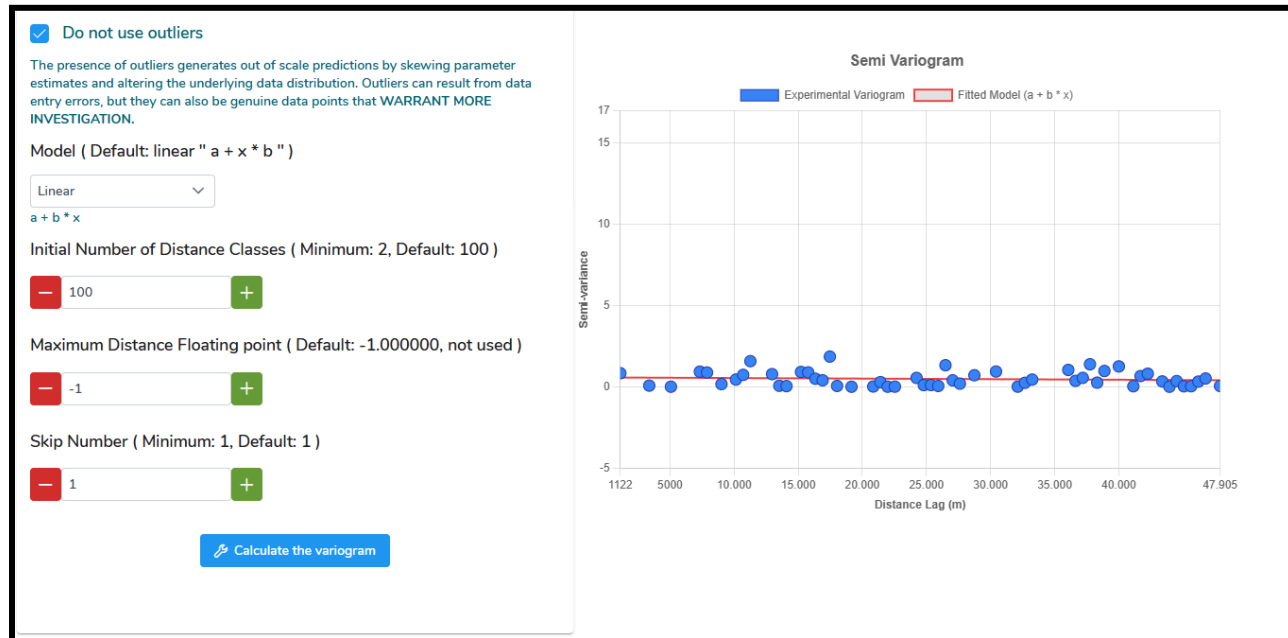


Figure 47 Publishing - Validation - Kriging Interpolation - Semi variogram

Dataset List

+ New Dataset

Clear

Search

Actions	Code	Name	Date	Context	Source	User	Email	Status
	6	admin-dataset-2026-09-17	2026-09-17		Sodium Exchangeable Percentage (ESP) sodicity	admin	demonitis@crs4.it	IN_PROCESS
	4	admin-dataset-2026-09-10	2026-09-10		Cation Exchange Capacity (CEC) / Clay ratio	admin	demonitis@crs4.it	CREATED

Figure 48 Publishing - Datasets in elaboration

Dataset List

+ New Dataset

Clear

Search

Actions	Code	Name	Date	Context	Source	User	Email	Status
	6	admin-dataset-2026-09-17	2026-09-17		Sodium Exchangeable Percentage (ESP) sodicity	admin	demonitis@crs4.it	PUBLISHED
	4	admin-dataset-2026-09-10	2026-09-10		Cation Exchange Capacity (CEC) / Clay ratio	admin	demonitis@crs4.it	CREATED

Figure 49 Publishing - Datasets published

After the elaboration the customized dataset will be published on the catalogue. The publishing report proposes a style (Figure 46) for the raster dataset because the generated raster dataset has a default style

Owner	demonitis@crs4.it
Date	2026-09-17
Source	Sodium Exchangeable Percentage (ESP) sodicity
Source points	0
Filtered points	28
Aggregated points	16
No results found	
Proposed Raster Style:	
<pre><?xml version="1.0" encoding="UTF-8"?> <StyledLayerDescriptor version="1.0.0" xsi:schemaLocation="http://www.opengis.net/sld StyledLayerDescriptor.xsd" xmlns="http://www.opengis.net/sld" xmlns:ogc="http://www.opengis.net/ogc" xmlns:xlink="http://www.w3.org/1999/xlink" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"> <NamedLayer> <Name>prediction_raster</Name> <UserStyle> <Title>Raster of predictions</Title> <Abstract>Raster of prediction</Abstract> <FeatureTypeStyle> <Rule> <Name>Prediction_raster</Name> <Title>Prediction_raster</Title> <RasterSymbolizer> <ChannelSelection> <GrayChannel> <SourceChannelName>1</SourceChannelName> </GrayChannel> </ChannelSelection> <ColorMap> <ColorMapEntry label="1.7533318996429443" quantity="1.7533318996429443" color="#d7191c"/> <ColorMapEntry label="1.8322239518165588" quantity="1.8322239518165588" color="#d7191c"/> <ColorMapEntry label="1.9111160039901733" quantity="1.9111160039901733" color="#d7191c"/> </ColorMap> </RasterSymbolizer> </Rule> </FeatureTypeStyle> </UserStyle> </NamedLayer> </StyledLayerDescriptor></pre>	

Figure 50 Publishing - Report

To change the default style of the raster open the dataset in the catalogue and then click on the “Edit” menu then “Edit style”

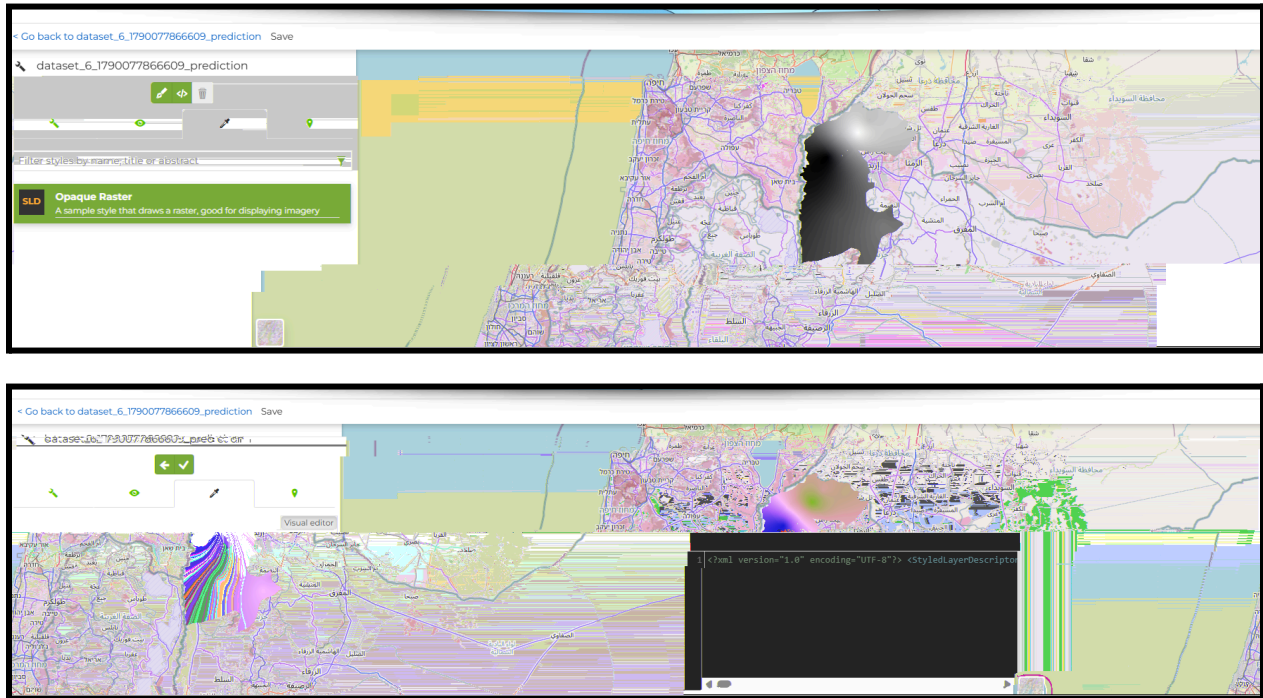


Figure 51 Publishing - changing raster style

Finally the data managers **must edit metadata and change the name** of the generated new datasets in the catalogue and then give permissions to the End User to view and download datasets

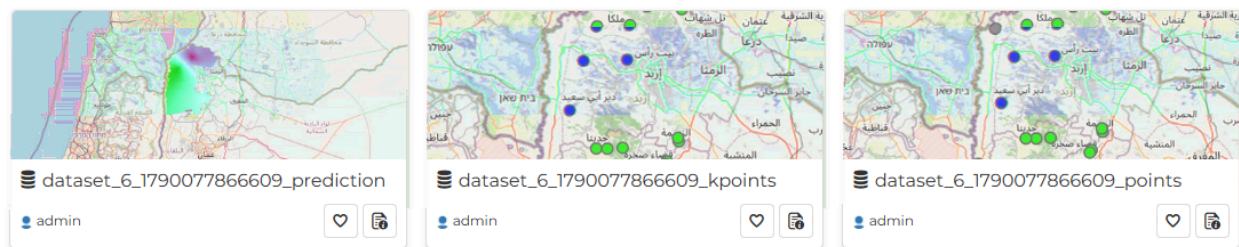


Figure 52 Publishing - generated rasters

The customized new dataset are:

- A raster with the interpolation (optional)
- A point vector dataset of aggregated points (the name end with '_kpoints') usable to perform a better interpolation in a desktop gis
- A point vector dataset of filtered points (the name end with '_points')

4. Admin Panel

The admin panel permits configuring the geonode catalogue. Only staff user can access to the panel

You can found a manual in the geonode documentation:

<https://docs.geonode.org/projects/v4/en/4.4.x/admin/index.html>

The screenshot displays the GeoNode administration interface. At the top, the header reads "GeoNode administration". Below this, the "Site administration" section is visible, containing several categories of administrative tasks, each with a list of items and "Add" and "Change" links.

ACCOUNTS

- Email addresses (+ Add, Change)

ACTIVITY STREAMS

- Actions (+ Add, Change)
- Follows (+ Add, Change)

ANNOUNCEMENTS

- Announcements (+ Add, Change)
- Dismissals (+ Add, Change)

ASSETS

- Local assets (+ Add, Change)

AUTHENTICATION AND AUTHORIZATION

- Groups (+ Add, Change)

AVATAR

- Avatars (+ Add, Change)

BACKOFFICE

- Base datasets (+ Add, Change)
- Climate and weathers (+ Add, Change)

Recent actions

My actions

- Soil Data Managers (Group profile)
- marta (User)
- muscas (User)
- muscas (User)
- roberto2 (User)
- Soil Data Managers (Group profile)
- belvisi (User)
- demontis (User)
- demontis (User)
- belvisi (User)

Figure 53 The Admin panel